

The University of Lahore
Department of Computer Science & IT
Midterm Examination



Spring-2022

Course Title:	Programming Fundamentals	Course Code:	CS02102	Credit Hours:	4
Course Instructors:	Dr. Abbas Khalid Mr. Muhammad Adress, Ms. Salwa Muhammad Akhtar, Ms. Naila Rafique	Program Name:	BSCS		
Time Allowed:	60 Minutes	Maximum Marks:	30		
Date:		Course Mentor:	Dr. Abbas Khalid		
Student's Name:		Signature:			
Reg. No:		Section:			

Important Instructions / Guidelines:

- Understanding of the questions is the part of the question paper so do not disturb the invigilators
- Make sure that you have 2 typed pages including this page.
- Kindly leave your mobile phones outside exam hall.
- **Solve question 1, 2 on question paper and question 3 on the answer sheet provided separately.**
- Your paper will be marked as ZERO, if involved in any kind of cheating.

Question#1: Choose the correct answer and encircle it. (5x1=5 Marks)

1. Which of the following has the highest precedence?

- a) + b) && c) || d) =

2. What will be the output of this code: `int i = 2, j = 3; cout<< j / i * j ;`

- a) 0 b) 1.8 c) 1 d) 3

3. Variable of float data type occupies _____ bytes in a memory.

- a) 2 b) 5 c) 6 d) 4

4. What will be the effect on a loop if a break statement is encountered?

- a) It immediately exits the loop
b) It skips a particular iteration and the loop continues
c) Loop counter is reset
d) None of these

5. Consider the following statements: `int a, x = 22, y=15; a = (x>y) ? (x+y) : (x-y);` What will be the value of a after executing these statements?

- a) 22 b) 37 c) 7 d) Error. Cannot be executed

CODE Segment	OUTPUT
<pre> int n = 1, i = 1; do{ cout << n << " "; while (i <= 10) { if (i < 5 && i != 2) cout <<"X"<<" "; i++; } }while (n++ <= 3); </pre>	1 X X X 2 3 4
<pre> int x , y=1; for (int i =0 ; i <= 5 ; i++) { cout<<"Enter number" ; cin>>x; if (a<= 0) continue; y *= x; } cout << y; Use x = 5, 10, -1, 3, -2 </pre>	OUTPUT: 5,10,3
<pre> int alpha = 9; switch (alpha % 6){ case 0: alpha--; cout<<alpha<<endl; case 1: case 2: alpha = alpha * 2; cout<<alpha<<endl; case 3: cout<<alpha<<endl; case 4: alpha = alpha - 5; cout<<alpha<<endl; case 5: alpha++; cout<<alpha<<endl; break; default: alpha = alpha / 3; cout<<alpha<<endl; } </pre>	OUTPUT: 9 4 5

1. Write a C++ program to convert a given integer (in seconds) to hours, minutes and seconds. **(5 marks)**

Solution:

```
#include <iostream>
int main() {
    int sec, h, m, s;
    cout<<"Input seconds: ";
    cin>>sec;

    h = (sec/3600);

    m = (sec -(3600*h))/60;

    s = (sec -(3600*h)-(m*60));

    cout<< "H:M:S - \n",h,m,s);

    return 0;
}
```

2. Write a C++ program which takes at least a two-digit number from user. Your program should calculate the sum of first and last digit of the input number. **(5 marks)**

Sample Output:

Input: 45
Output: 9

Sample Output:

Input: 456
Output: 10

Sample Output:

Input: 6786
Output: 12

Solution:

```
#include <iostream>
using namespace std;
int main()
{
    int num, sum=0, firstDigit, lastDigit;
```

```
cout<<"Enter any number:";

cin>>num;

lastDigit = num % 10;

firstDigit = num;

while(num >= 10)

{

num = num / 10;

}

firstDigit = num;

sum = firstDigit + lastDigit;

cout<<"Sum of first and last digit: "<<sum;

return 0;

}
```